

Week 2: Building Intuition

Jacob Khohayting

Last updated: September 20, 2026

Contents

Making Visual Observations	2
Patterns: Warm-Up	3
Repeating Patterns	4
Number Sequences	5
Visual Patterns	6
Practice Problems	7

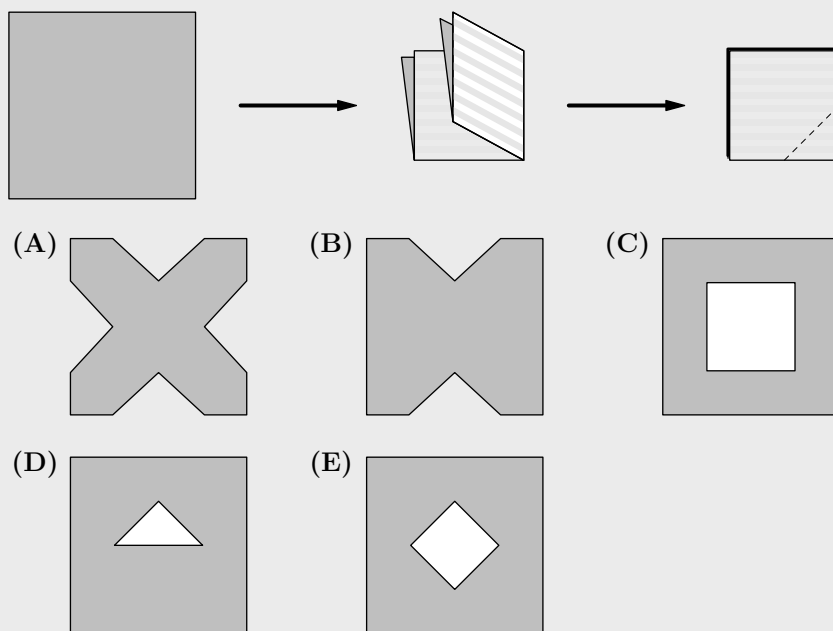
※ Making Visual Observations

Strong problem solvers look before they calculate. They notice symmetry, recognize patterns, and mentally transform figures to predict what will happen. The problems in this section are designed to sharpen these visual reasoning skills, which appear frequently in mathematical competitions.

Before reaching for algebra or arithmetic, spend a moment simply observing. Can you spot a pattern? Is there symmetry? Can you imagine what happens after a fold, a rotation, or a rearrangement? Developing this habit of making visual observations will help you solve many competition problems more quickly and elegantly.

In general, always feel free to make small notes for yourself and *write stuff down*. It is hard to visualize things and so feel free to make whatever notes that will help you!

A square piece of paper is folded twice into four equal quarters, as shown below, then cut along the dashed line. When unfolded, the paper will match which of the following figures?



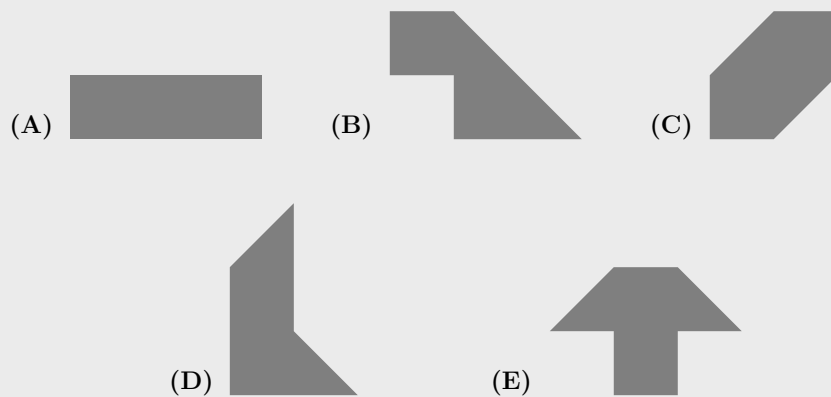
Split the paper into 4 quadrants.

By symmetry (which is present because of the double-fold), the hole in the unfolded paper will be made of 4 triangles (each of which is congruent to the one cut out), so the unfolded figure will match the figure in E.

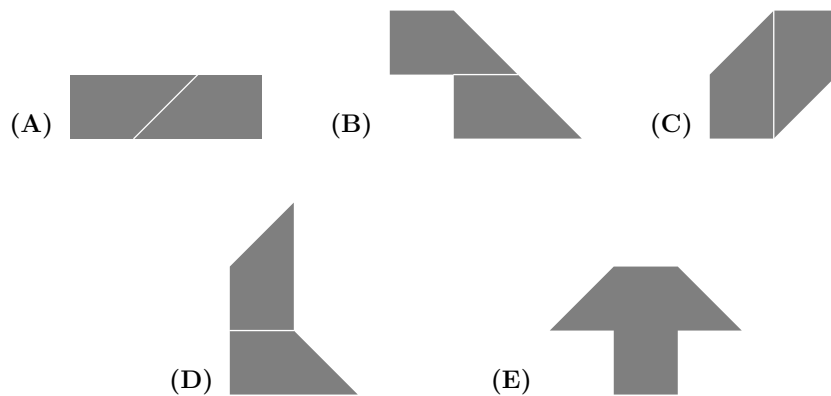
Eva assembles these two identical parts into a figure.



Which of these figures can she not build?



Here are each of the constructions:



This leaves us with E, which is the correct answer. Sometimes it is easier to prove other choices are wrong and use the process of elimination to be sure!

We will see more of these problems later. They are all about getting used to.

✂ Patterns: Warm-Up

What is the 99th shape in the sequence?

$\triangle, \square, \square, \diamond, \triangle, \square, \square, \diamond, \triangle, \square, \square, \diamond, \triangle, \square.$

Once you got this, answer the following questions?

- Did you draw all 99 shapes?
- Did you notice something repeating?
- If so, what was repeating?
- What is the length of the repeating section?

Solution. By observing the sequence, we can see that the pattern consists of 4 repeating shapes:

$$\triangle, \square, \square, \diamond.$$

Since the pattern repeats every 4 terms, we can determine the 99th shape by finding the remainder when 99 is divided by 4: $99 \div 4 = 24$ with a remainder of 3.

A remainder of 1 corresponds to the first shape ($\triangle$), a remainder of 2 corresponds to the second shape ($\square$), a remainder of 3 corresponds to the third shape ($\square$), and a remainder of 0 corresponds to the fourth shape ($\diamond$). Since the remainder is 3, the 99th shape is the third shape in the repeating sequence.

Therefore, the 99th shape is: $\square$.

※ Repeating Patterns

Here is the process, we will use on these types of problems. Retry this on the first problem to get the hang of it first!

- Circle the smallest repeating group.
- Count how many items are in one group.
- Divide or skip-count to get close to the desired number.
- Find the leftover position.

Mistake: Students start counting from the beginning every time. That works for small numbers, but contest problems often ask for the 25th, 50th, or 100th item.

Here, is a similar problem to ensure we are on the same page!

A bracelet has beads in this repeating pattern:

Red, Blue, Blue, Green, Red, Blue, Blue, Green, ...

What color is the 38th bead?

※ Number Sequences

Do not just look at the numbers. Look at how the numbers change. As we will see, the pattern won't always be obvious. You will have to inspect them sometimes to see what is really going on.

What is the next term of the sequence 3, 6, 9, 12, 15, __ ? Can you write the next five terms?

Solution. If you look closely at the sequence 3, 6, 9, 12, 15, . . . , the underlying pattern jumps out pretty quickly. This is a classic arithmetic progression where every single term increases by exactly 3 from the one before it.

In mathematical terms, we can describe the general n -th term with the straightforward formula $a_n = 3n$, which is essentially just walking right through the multiples of 3.

Since the given list stops at 15 (which is the 5th term), finding the immediate next term is a breeze. We just need to add 3 to that last number, or plug 6 into our formula to get $a_6 = 3 \times 6 = 18$. So, 18 is the next number up.

To stretch that out for the next five terms, we just keep the momentum going by adding 3 to each successive result. Doing that math rolls us smoothly from 18 to 21, then 24, 27, and finally landing on 30. Written out altogether, the next five terms of your sequence are:

$$18, 21, 24, 27, 30.$$

What is the next term of the sequence 2, 5, 10, 17, 26, __ ? Can you write the next five terms?

Solution. To find the next terms in the sequence 2, 5, 10, 17, 26, . . . , it helps to look at how much the value increases between each step:

- $5 - 2 = 3$
- $10 - 5 = 5$
- $17 - 10 = 7$
- $26 - 17 = 9$

You can see that the difference between terms is growing by 2 each time (3, 5, 7, 9, . . .). This pattern suggests we are dealing with consecutive odd numbers being added to each term.

Another way to look at this is by recognizing that each term is actually one greater than a perfect square: $1^2 + 1 = 2$, $2^2 + 1 = 5$, $3^2 + 1 = 10$, $4^2 + 1 = 17$, and $5^2 + 1 = 26$. Using the formula $a_n = n^2 + 1$, we can easily find the next steps. Following the pattern of adding the next odd number (11, 13, 15, 17, 19), the next term is $26 + 11 = 37$. Continuing this logic for the next five terms, we get:

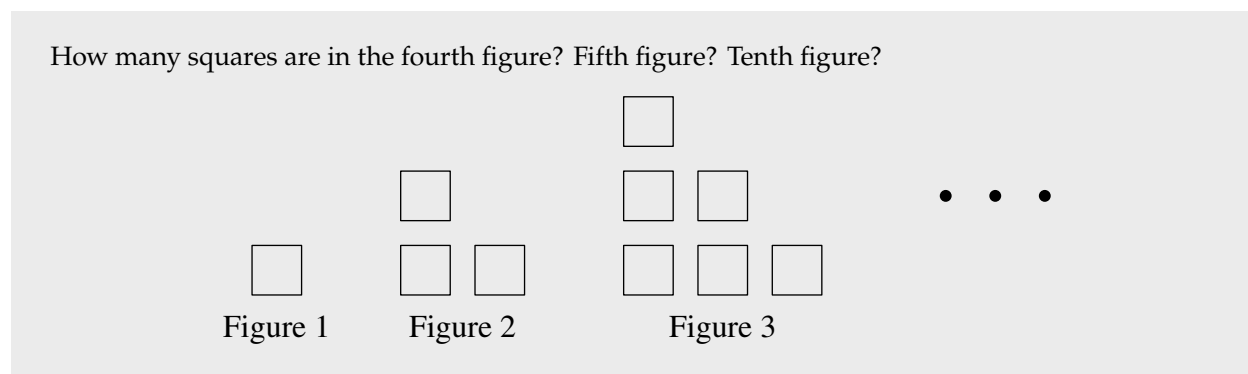
- $6^2 + 1 = 36 + 1 = 37$
- $7^2 + 1 = 49 + 1 = 50$

- $8^2 + 1 = 64 + 1 = 65$
- $9^2 + 1 = 81 + 1 = 82$
- $10^2 + 1 = 100 + 1 = 101$

So, the next five terms in your sequence are:

37, 50, 65, 82, 101.

※ Visual Patterns



For these problems, it is always a good idea to draw a table. Sometimes, it is hard to understand things visually. Other times, it helps. Here is a chart to help you start:

Figure 1	1
Figure 2	3
Figure 3	6
Figure 4	—
Figure 5	—
Figure 6	—

Solution. If you look closely at how these figures grow, you can see a really neat pattern hiding in plain sight. Figure 1 starts with just 1 square. To build Figure 2, you keep that first square and slide a new row of 2 squares underneath it, giving you a total of 3. For Figure 3, you add a row of 3 squares to the bottom, bringing the total to 6. Every time you move to the next figure, you are just adding a new bottom row that has the exact same number of squares as the figure number itself.

Following this logic makes finding the next few shapes a breeze without needing to draw them. To get the fourth figure, you just take the 6 squares you already have and add a new row of 4 squares to the bottom, which lands you at exactly 10 squares. For the fifth figure, you keep the momentum going by adding a row of 5 squares to that previous total, giving you 15 squares.

Now, finding the tenth figure might seem like a lot of work if you try to count it all out, but since we know the secret, we just need to add up all the numbers from 1 to 10. A classic trick for MOEMS competitions is to pair the numbers up from the outside in to save time. If you match 1 with 10, 2 with 9, 3 with 8, 4 with 7, and 5 with 6, you end up with five pairs that each add up to 11. Multiplying those 5 pairs by 11 gives you a grand total of 55 squares for the tenth figure, completely bypassing the tedious counting!

※ Practice Problems

1. Here are some tougher sequences. Can you fill in the rest of the terms?

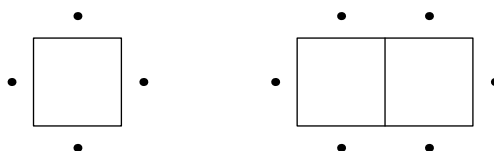
- 1, 2, 3, 5, 8, 13, __, __, __, ...
- 2, 4, 8, 16, __, __, __, ...
- 10, 20, 15, 25, 20, 30, __, __, __, ...

2. What is the next 34th letter of the sequence

A, B, C, C, D, A, B, C, C, D, ...

3. If we count by 3s starting with 1, the following sequence is obtained: 1, 4, 7, 10, What is the 100th number in the sequence?

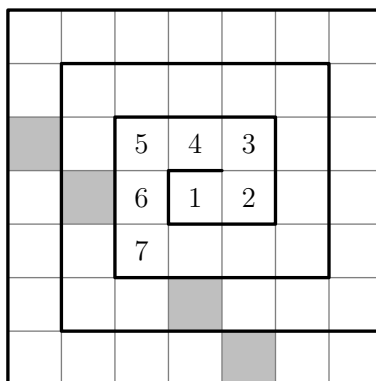
4. Tables are pushed together as follows:



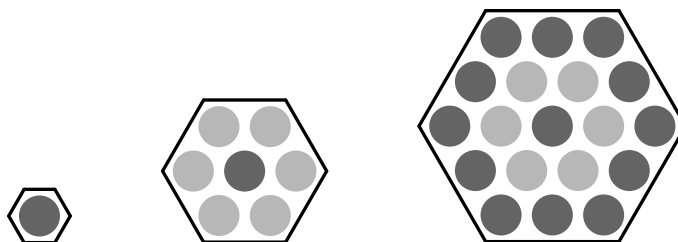
Above shows the cases for one table and two tables. How many people could be seated if there were 10 tables? How about 20 tables?

5. Lucius is counting backwards by 7s. His first three numbers are 100, 93, and 86. What is his 10th number?

6. The numbers from 1 to 49 are arranged in a spiral pattern on a square grid, beginning at the center. The first few numbers have been entered into the grid below. Consider the four numbers that will appear in the shaded squares, on the same diagonal as the number 7. How many of these four numbers are prime?



7. Three hexagons of increasing size are shown below. Suppose the dot pattern continues so that each successive hexagon contains one more band of dots. How many dots are in the next hexagon?



8. The counting numbers are arranged in four columns, as shown below. Under which column letter will 101 appear?

A	B	C	D
1	2	3	4
8	7	6	5
9	10	11	12
...	...	14	13